Vikas Purbia

AI Engineer | Machine Learning | Generative AI

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LinkedIn | GitHub

Professional Summary

B.Tech student in Artificial Intelligence & Data Science with hands-on projects in RAG applications, machine learning, computer vision, NLP, and customer analytics. Experienced with Python, model evaluation, hybrid retrieval, clustering, and building interactive applications. Focused on developing reliable AI systems and validating performance with measurable evaluation results.

Technical Skills

Languages: Python, SQL
ML/DL: Scikit-learn, XGBoost, TensorFlow, Keras, CNNs, transfer learning, K-Means, model evaluation
Generative AI: RAG, LLM applications, hybrid retrieval, BM25, dense-vector search, Reciprocal Rank Fusion
Data & NLP: Pandas, NumPy, TF-IDF, sentiment analysis, RFM analysis, cosine similarity
Tools & Deployment: FastAPI, Streamlit, Ollama, Qdrant, SQLite/MySQL, Git, GitHub, Jupyter Notebook

Selected Projects

Support Copilot – AI-Powered RAG Support Assistant GitHub
Python, FastAPI, Ollama, Qdrant, Streamlit

- Built a local RAG support assistant with source citations, role-based document filtering, confidence checks, and human escalation.
- Combined dense-vector retrieval and BM25 using Reciprocal Rank Fusion; evaluated 11 answerable and 3 unanswerable cases.
- Test results: answered 10/11 answerable questions, cited the correct source in 10/11, and correctly refused 3/3 unanswerable questions.

Integrated Retail Analytics for Store Optimization GitHub
Python, XGBoost, Pandas, scikit-learn

- Developed a weekly sales-forecasting workflow using historical sales, store attributes, promotions, and economic indicators.
- Project evaluation reported overall MAE of 1,357.57 and RMSE of 3,018.71; explored store/department-level error variation, anomaly detection, segmentation, and market-basket analysis.

Shopper Spectrum – Customer Segmentation & Product Recommendations GitHub
Python, Pandas, scikit-learn, Streamlit

- Prepared 541,909 raw transaction rows and retained 397,884 after filtering missing customer IDs, cancellations, and non-positive quantity/price records.
- Engineered RFM features and evaluated K-Means for $k = 2$–10; the supplied evaluation plot showed a best silhouette score of approximately 0.617 at $k = 5$.
- Built a cosine-similarity-based product recommendation workflow and an interactive Streamlit interface.

Multiclass Fish Image Classification GitHub
Python, TensorFlow, Keras, MobileNetV2

- Developed a 12-class image-classification workflow using a custom CNN and MobileNetV2 transfer learning.
- Fine-tuned the final 20 MobileNetV2 layers; recorded 98.17% validation accuracy and 0.0495 validation loss after three fine-tuning epochs. These are validation, not held-out test, metrics.

Zomato Restaurant Clustering & Sentiment Analysis GitHub

- Analyzed 10,000 reviews across 100 restaurants; applied TF-IDF text features (17,333 features) for sentiment-classification experiments.
- Evaluated K-Means cluster counts from $k = 2$ to 9; best saved silhouette score was 0.325 at $k = 6$. Implemented Logistic Regression and XGBoost classification workflows.

Experience

AI/ML Intern – Labmentix Pvt. Ltd. Bengaluru, India
Jul 2025 – Sep 2025

- Worked on Python-based data preparation, analysis, and machine-learning workflows.
- Used analytical and ML tools to support project tasks and communicate data findings.

Education

B.Tech, Artificial Intelligence & Data Science 2023 – 2027
Bikaner Technical University CGPA: 8.04/10